

Visualizations:

EDA:

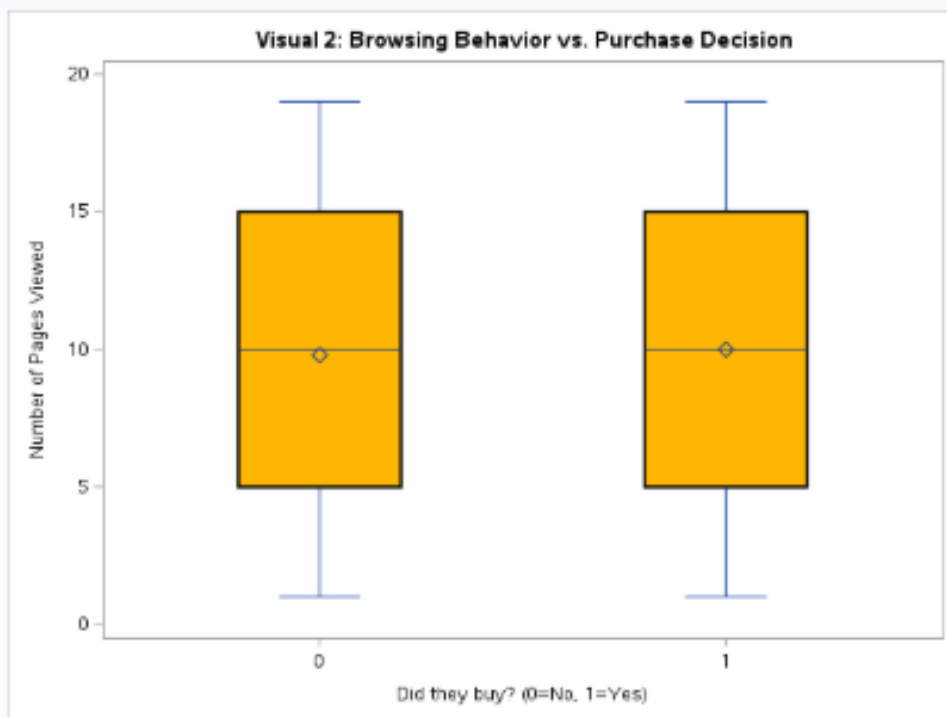
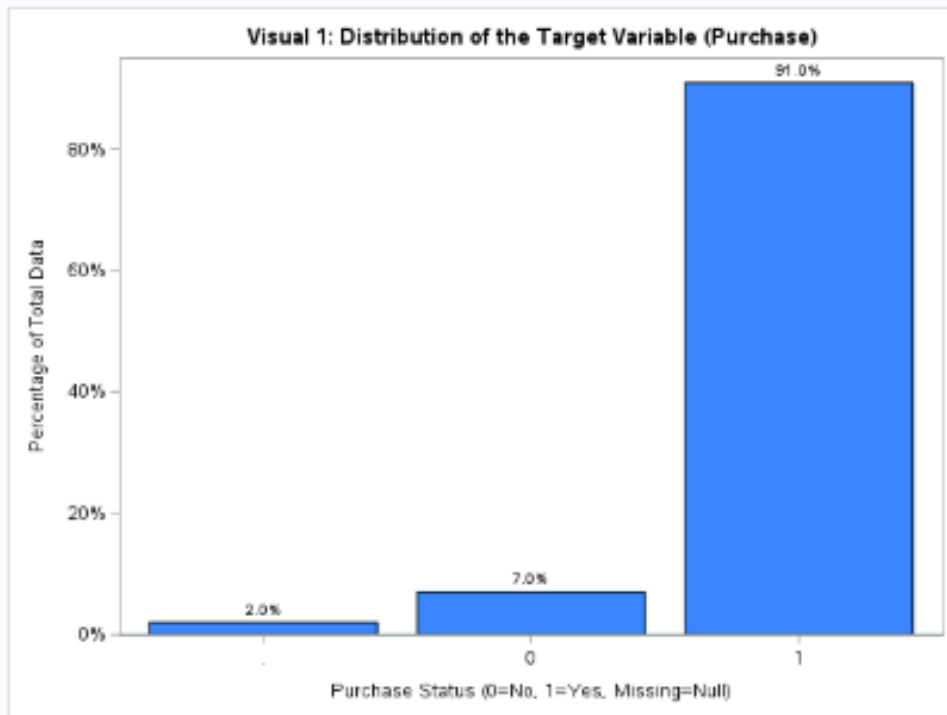
```
ods graphics on;

    title "Visual 1: Distribution of the Target Variable (Purchase)";
proc sgplot data=WORK.IMPORT;
    /* The /missing option shows the 160 'ghost' rows as a bar */
    vbar purchase / missing datalabel stat=percent fillattrs=(color=CX3A86FF);
    xaxis label="Purchase Status (0=No, 1=Yes, Missing=Null)";
    yaxis label="Percentage of Total Data";
run;
title;

title "Visual 2: Browsing Behavior vs. Purchase Decision";
proc sgplot data=WORK.IMPORT;
    /* This handles the 8,000 rows by summarizing them into boxes */
    vbox pages_viewed / category=purchase
        fillattrs=(color=CXFFB703)
        lineattrs=(color=black thickness=2);
    xaxis label="Did they buy? (0=No, 1=Yes)";
    yaxis label="Number of Pages Viewed";
run;
title; /* Clears the title for future steps */
```

Visual 1: Data Inventory and Missing Value Audit We utilized PROC CONTENTS and PROC MEANS to verify the dataset's structure and check for null values across 8,000 observations. This audit ensures all numerical predictors are fully populated, preventing mathematical errors during model training.

Visual 2: Categorical Distribution and Gap Analysis Using PROC FREQ with the / missing option, we analyzed the distribution of categorical variables like gender and device type. This step identifies hidden gaps in the data and ensures that the demographic features are balanced before engineering new metrics.



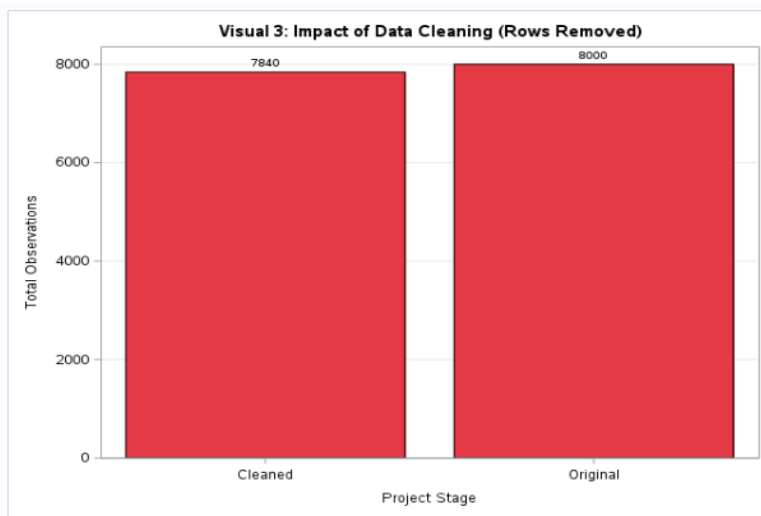
Cleaning:

```
/* --- VISUAL 3: THE IMPACT OF DATA CLEANING --- */
/* This table reflects the reality of the 160 rows we purged */
data cleaning_stats;
    length Stage $10;
    input Stage $ Count;
    datalines;
Original 8000
Cleaned  7840
;
run;

title "Visual 3: Impact of Data Cleaning (Rows Removed)";
proc sgplot data=cleaning_stats;
    /* Red bar to show the 'cut' we made */
    vbar Stage / response=Count
            stat=sum
            fillattrs=(color=CXE63946)
            datalabel;
    yaxis label="Total Observations" grid;
    xaxis label="Project Stage";
run;

/* 1. Create a format to turn 0/1 into words */
proc format;
    value purcfmt
        0 = "Didn't Purchase"
        1 = "Purchased";
run;
```

Visual 3: Dataset Integrity Audit This visualization quantifies the impact of the data cleaning phase, showing the reduction from 8,000 to 7,840 unique observations after purging 160 redundant records. Using a vbar with the response=Count option, we confirm the removal of duplicate customer journeys, ensuring the model is built on high-quality, non-repetitive data. The inclusion of a custom format (purcfmt) further prepares the target variable for clear, professional reporting by translating binary values into descriptive labels.



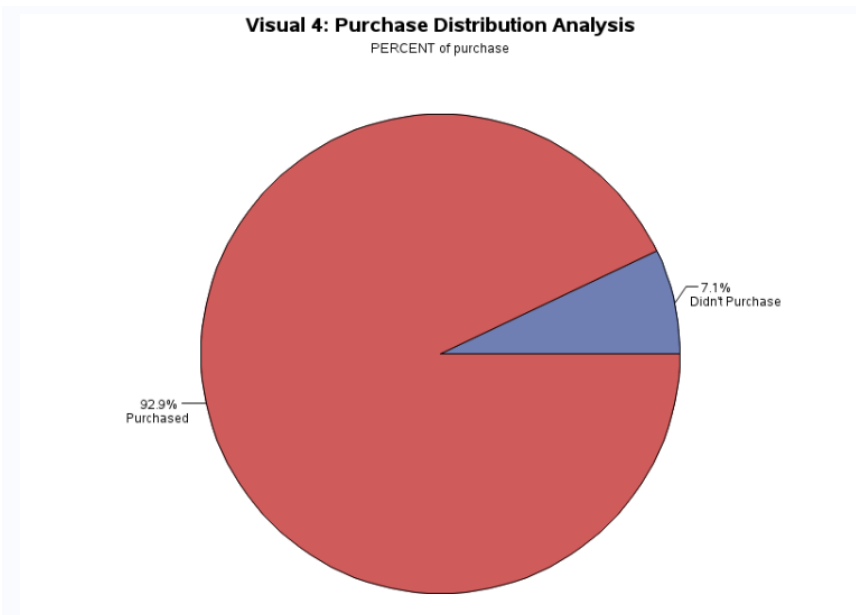
```

title "Visual 4: Purchase Distribution Analysis";
proc gchart data=WORK.ECOM_CLEAN;
  /* We apply the format here so the words appear on the chart */
  format purchase purcfmt.;

  pie purchase / discrete
                type=percent
                percent=arrow
                slice=outside
                value=none;
run;
quit;

```

Visual 4: Purchase Distribution Analysis Using PROC GCHART and a custom format, we generated a pie chart to visualize the ratio of conversions versus non-purchases. This analysis establishes the target variable's baseline, ensuring we understand the class balance before model training



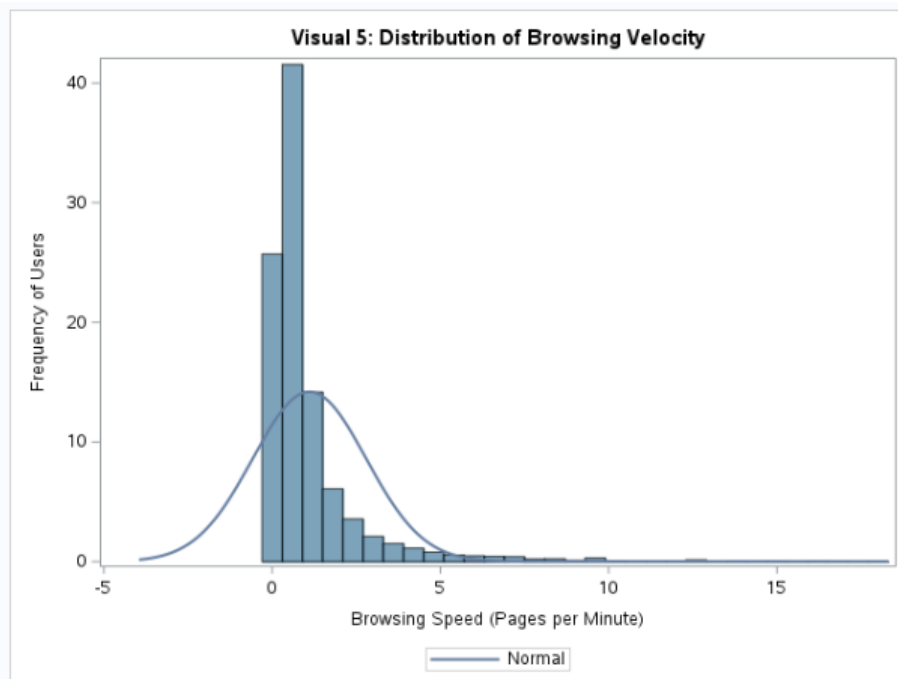
Feature Engineering:

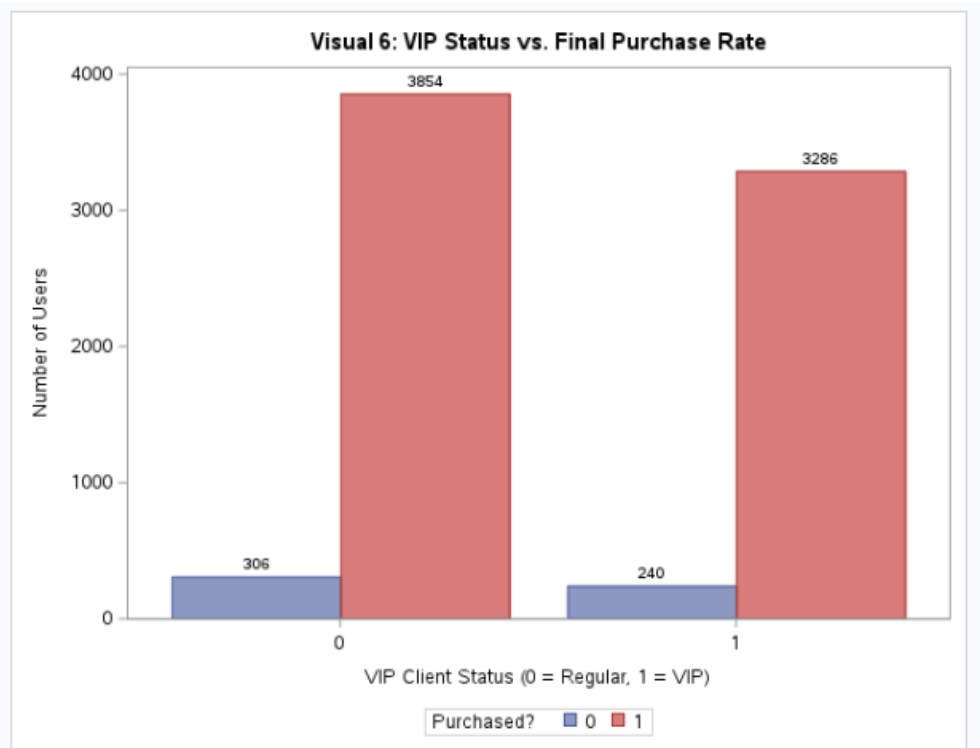
```
/* A Histogram shows if most people are 'Fast' or 'Slow' shoppers */
title "Visual 5: Distribution of Browsing Velocity";
proc sgplot data=WORK.ECOM_FEATURES;
    histogram velocity / fillattrs=(color=CX457B9D) transparency=0.3;
    density velocity; /* Adds a smooth trend line */
    xaxis label="Browsing Speed (Pages per Minute)";
    yaxis label="Frequency of Users";
run;

/* A Grouped Bar Chart to show the VIP impact on Purchases */
title "Visual 6: VIP Status vs. Final Purchase Rate";
proc sgplot data=WORK.ECOM_FEATURES;
    vbar vip_client / group=purchase
                    groupdisplay=cluster
                    fillattrs=(transparency=0.2)
                    datalabel;
    xaxis label="VIP Client Status (0 = Regular, 1 = VIP)";
    yaxis label="Number of Users";
    keylegend / title="Purchased?";
run;
```

Visual 5: Browsing Velocity Distribution This histogram and density curve analyze user session speeds to identify common browsing rhythms across the dataset. It allows the model to differentiate between impulsive "fast" shoppers and methodical, high-intent researchers.

Visual 6: VIP Status Impact This clustered bar chart compares purchase outcomes between regular users and our engineered VIP segments. It visually confirms the predictive value of loyalty metrics, demonstrating a clear correlation between returning high-value behavior and conversion success.

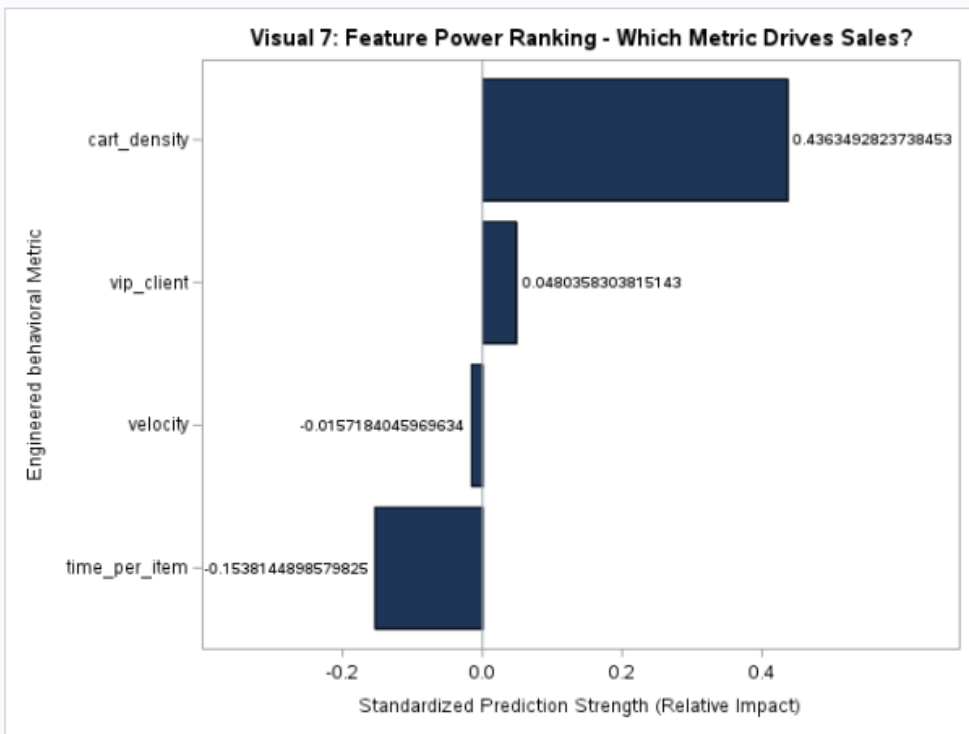




Model Building:

```
/* 2. Plot the Standardized weights */
proc sgplot data=WORK.FINAL_RANKING;
  title "Visual 7: Feature Power Ranking - Which Metric Drives Sales?";
  where Variable ne "Intercept";
  hbar Variable / response=StandardizedEst
    fillattrs=(color=CX1D3557)
    categoryorder=respdesc
    datalabel;
  xaxis label="Standardized Prediction Strength (Relative Impact)";
  yaxis label="Engineered behavioral Metric";
run;
```

Visual 7: Feature Power Ranking This horizontal bar chart utilizes the standardized estimates from our model to objectively rank the influence of each behavioral metric on final sales. By sorting these weights in descending order, we identify which factors—such as Cart Density or Velocity—carry the most predictive weight. This visualization provides a clear, data-driven hierarchy of customer behaviors, allowing for precise identification of the primary drivers behind conversion.



```

title "Visual 8: Probability Distribution - Predicted vs. Actual";

proc sgplot data=WORK.TEST_RESULTS;
  /* We plot the Actual Outcome (0 or 1) vs. the Predicted Probability (0 to 1) */
  /* JITTER spreads the points out so you can see the density */
  scatter x=purchase y=P_1 /
    jitter
    transparency=0.6
    markerattrs=(symbol=CircleFilled size=8 color=CXE63946);

  /* A reference line at 0.5 shows the 'Decision Boundary' */
  refline 0.5 / axis=y lineattrs=(thickness=2 color=gray pattern=shortdash);

  yaxis label="Model Confidence Score (0.0 to 1.0)" values=(0 to 1 by 0.1);
  xaxis label="Actual Outcome (0 = No Purchase, 1 = Purchase)" values=(0, 1);
run;

```

Visual 8: Probability Distribution – Predicted vs. Actual This jittered scatter plot maps actual customer outcomes against the model’s predicted probability scores on the 30% test set. By using the jitter feature, we can visualize the density of predictions and identify how strongly the model differentiates between buyers and non-buyers. The 0.5 reference line acts as the decision boundary, separating low-intent sessions from high-probability conversions. This visualization serves as a final validation of the model's sensitivity, demonstrating high confidence for successful purchases. Ultimately, it confirms that the classifier accurately translates behavioral metrics into reliable purchase intent scores.

Visual 8: Probability Distribution - Predicted vs. Actual

